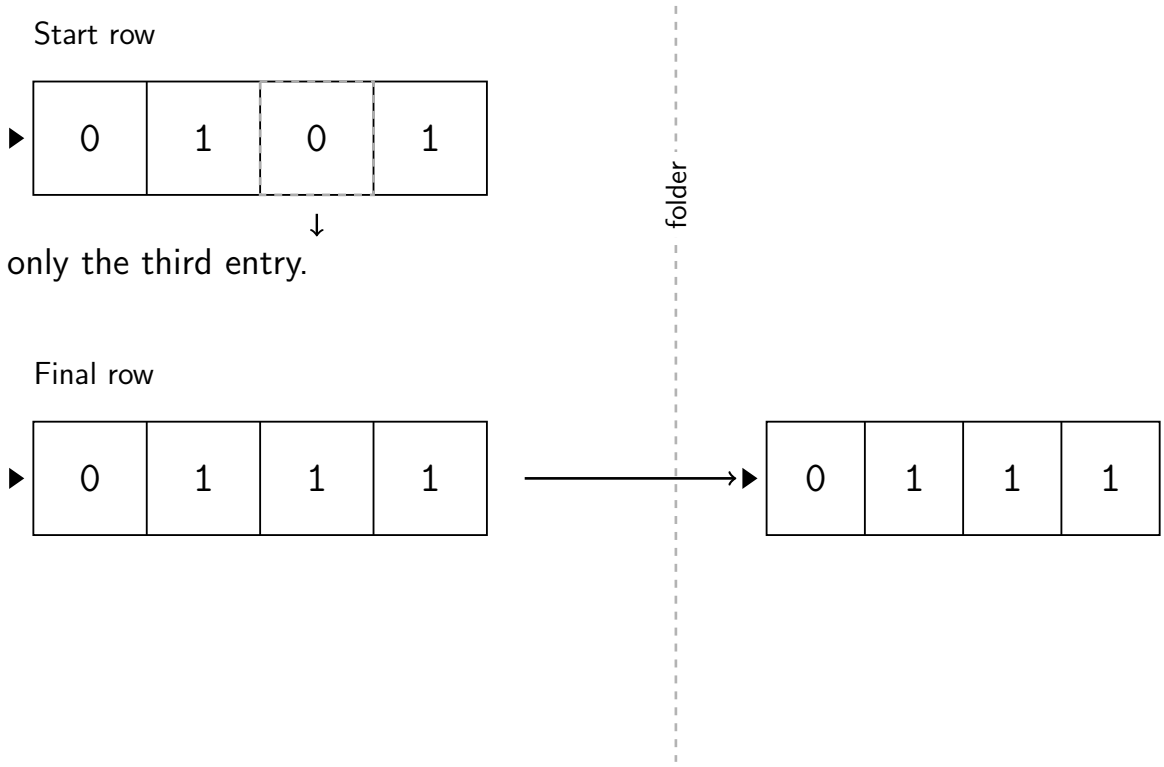
Use 0 for an empty circle and 1 for a filled circle. A key gives a different row to each picture, all the same length. The sender hides the picture and its row. A changer changes zero or one entry, without saying where or whether a change occurred. The receiver gets only the final row and knows the key. Keep the arrow at the left.

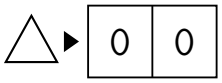
Sender and changer

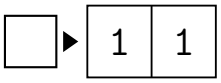
Receiver

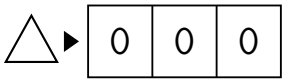


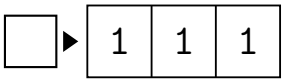
In the game, the start row and the changed position stay hidden behind the folder.

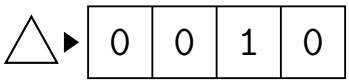
Problem 1: For each key, find every row the receiver might see. Mark any row that could mean either picture. Which keys always let the receiver recover the picture?

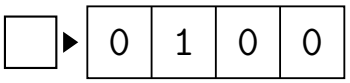




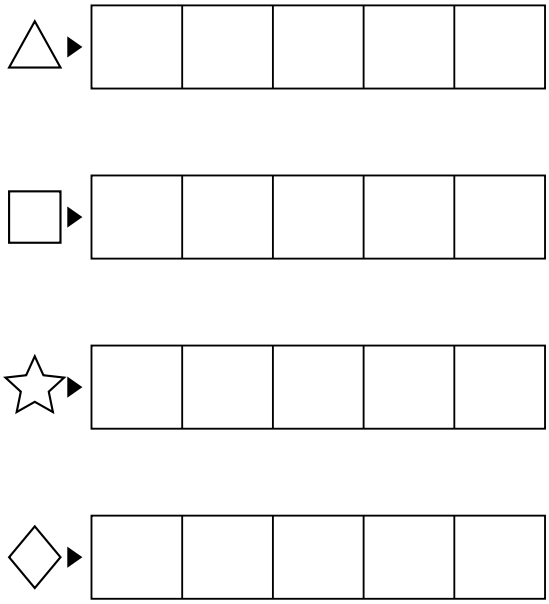






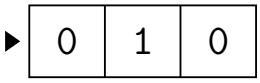


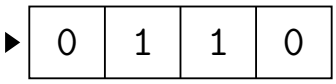
Problem 2: Make a key for all four pictures, with five entries in each row, that always lets the receiver recover the picture. Explain why your key works.

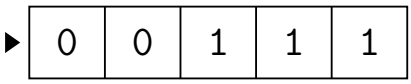


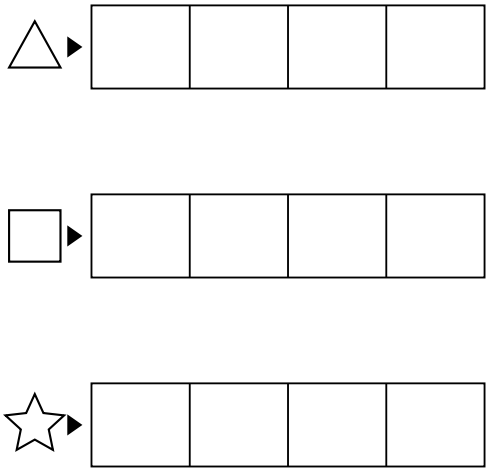
Problem 3: For each pair, could the receiver see the same row after either picture was sent? What is the smallest number of differing positions that keeps two pictures apart for rows of any length? Explain why that number is enough and why a smaller number fails.

Problem 4: Find every row the receiver might see from each row below, and explain why none are missing. How many received rows can come from any one row of length 10?







Problem 5: What is the shortest row length for a key that always works for four pictures? Give a working key and explain why every shorter length is impossible.

Problem 6: Can three pictures have a key that always works with four entries in each row? Find one, or explain why every choice fails.